

Practical # 10

Write a program that takes input a number and then check it is prime or not.

```
3 ~ int main() {
4     int num, i, isPrime = 1;
5
6     printf("Enter a number: ");
7     scanf("%d", &num);
8
9 ~     if (num <= 1) {
10         isPrime = 0;
11 ~     } else {
12 ~         for (i = 2; i <= num / 2; i++) {
13 ~             if (num % i == 0) {
14                 isPrime = 0;
15                 break;
16             }
17         }
18     }
19
20     if (isPrime)
21         printf("The number is Prime.\n");
22     else
23         printf("The number is Not Prime.\n");
24
25     return 0;
26 }
```

```
Enter a number: 2
The number is Prime.
```

Algorithm:

1. Start
2. Declare integer variables for the number and loop counter
3. Read the number from the user
4. If the number is less than or equal to 1, mark it as not prime
5. Otherwise, check divisibility from 2 to half of the number
6. If divisible by any value, mark it as not prime and stop checking
7. Display whether the number is prime or not
8. End